Geometry Right Triangles & Trigonometry Cheat Sheet

Unit Focus: the Pythagorean theorem and special right triangles

1 RIGHT TRIANGLE VOCABULARY

Right angle = exactly 90°.

Legs = the sides forming the 90° angle.

Hypotenuse = opposite the 90° angle.

Hypotenuse is the longest side.

Pythagorean theorem: $a^2 + b^2 = c^2$.

Use c ONLY for the hypotenuse.

2 PYTHAGOREAN EXAMPLE

Find c if the legs are 6 and 8.

$a^2 + b^2 = c^2$

$6^2 + 8^2 = c^2$

$36 + 64 = c^2$

$100 = c^2$

$c = 10$ units

3 PYTHAGOREAN CONVERSE

For positive side lengths a, b, c,

with c the longest, if

$a^2 + b^2 = c^2$, the triangle is right.

Check 5, 12, 13:

$5^2 + 12^2 = 25 + 144 = 169$.

$13^2 = 169$. Therefore it is right.

4 45-45-90 TRIANGLES

Two 45° angles; two equal legs.

Side ratio: x : x : x sqrt(2)

(leg : leg : hypotenuse)

If each leg is 7,

hypotenuse = 7 sqrt(2).

For hypotenuse 10 sqrt(2), leg = 10.

5 30-60-90 TRIANGLES

Side ratio: x : x sqrt(3) : 2x

(short leg : long leg : hypotenuse)

Short leg is opposite 30°.

Long leg is opposite 60°.

Example: short leg = 5

long leg = 5 sqrt(3)

hypotenuse = 10.

6 FIND A MISSING LEG

Hypotenuse = 13; one leg = 5.

$5^2 + b^2 = 13^2$

$25 + b^2 = 169$

$b^2 = 144$

b = 12 (take positive root)

Check: 25 + 144 = 169.

7 QUICK REMINDERS & COMMON MISTAKES

✓ Identify the hypotenuse BEFORE using $a^2 + b^2 = c^2$.

✓ Special right triangle ratios require 45-45-90 or 30-60-90 angles.

✗ Never add leg lengths to find a hypotenuse. ✗ Never use a leg as c.

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More Practice: SOH-CAH-TOA and inverse trig

1 TRIG RATIOS

In a right triangle, the trigonometric ratios relate an angle to the lengths of the sides.

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$$

Memory Aid:

SOH - CAH - TOA

Sine = Opposite / Hypotenuse

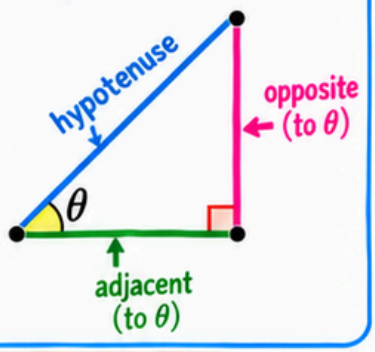
Cosine = Adjacent / Hypotenuse

Tangent = Opposite / Adjacent

2 LABELING SIDES

The names of the sides (opposite, adjacent, hypotenuse) depend on which angle you are using.

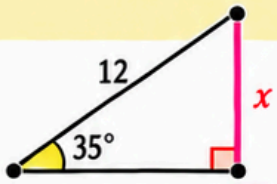
- **Opposite side:** across from the chosen angle (θ).
- **Adjacent side:** next to the chosen angle, but not the hypotenuse.
- **Hypotenuse:** across from the right angle. It is always the longest side.



3 FINDING A SIDE WITH SINE

Use sine when you know the **opposite** side and the hypotenuse (or need to find the **opposite** side).

Example: Find the length of x.



Step 1: Write the trig ratio.

$$\sin 35° = \frac{x}{12}$$

Step 2: Solve for x.

$$x = 12\sin 35°$$

Step 3: Use a calculator.

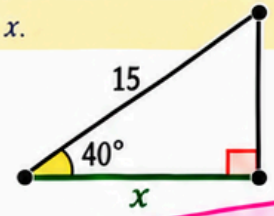
$$x \approx 6.9$$

Answer: $x \approx 6.9$

4 FINDING A SIDE WITH COSINE

Use cosine when you know the **adjacent side** and the hypotenuse (or need to find the **adjacent** side).

Example: Find the length of x.



Step 1: Write the trig ratio.

$$\cos 40° = \frac{x}{15}$$

Step 2: Solve for x.

$$x = 15\cos 40°$$

Step 3: Use a calculator.

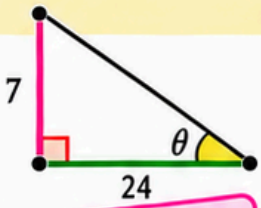
$$x \approx 11.5$$

Answer: $x \approx 11.5$

5 FINDING AN ANGLE WITH INVERSE TANGENT

Use tangent when you know the **opposite** and **adjacent** sides and need to find the angle.

Example: Find the measure of θ.



Step 1: Write the trig ratio.

$$\tan\theta = \frac{7}{24}$$

Step 2: Solve for θ using inverse tangent.

$$\theta = \tan^{-1}\left(\frac{7}{24}\right)$$

Step 3: Use a calculator.

$$\theta \approx 16.3°$$

Answer: $\theta \approx 16.3°$

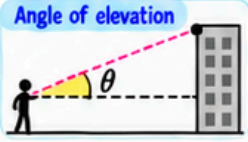
6 CHOOSING THE CORRECT RATIO

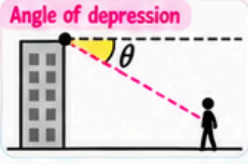
Pick the trig ratio that matches the sides you know (or the side you need to find).

Known Sides	Use This Ratio	
Opposite and hypotenuse	**Sine**	$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$
Adjacent and hypotenuse	**Cosine**	$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$
Opposite and adjacent	**Tangent**	$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$

7 ANGLE OF ELEVATION / DEPRESSION

- **Angle of elevation:** the angle formed by a line of sight that looks upward from a horizontal line.
- **Angle of depression:** the angle formed by a line of sight that looks downward from a horizontal line.





The trigonometric work is the same for both. Draw and label the diagram correctly, identify the angle and the sides, then use the appropriate trig ratio (SOH-CAH-TOA).

8 QUICK STRATEGY TIPS & COMMON MISTAKES

QUICK TIPS:

- Set your calculator to degree mode (DEG).
- Draw and label the right triangle.
- Identify the sides (opposite, adjacent, hypotenuse) relative to the chosen angle before selecting a trig ratio.
- Round your final answer as directed (e.g., nearest tenth).
- Include units when appropriate (e.g., cm, m, ft, degrees).

COMMON MISTAKES:

- Using the wrong trig ratio for the given sides.
- Mixing up opposite and adjaceet sides.
- Not setting the calculater to degree mode.
- Rounding too early in multi-step problems.
- Forgetting to draw and label a diagram.

Back